

Sean Luka Girgis

Senior Performance Test Lead / Performance Engineer

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Professional Summary

Senior Performance Test Lead and Performance Engineer with 20+ years of enterprise IT experience across performance testing, APM/observability, Java/J2EE diagnostics, telemetry analysis, capacity planning, and stakeholder reporting. Strong background in JMeter-style performance testing, LoadRunner/Performance Center, Dynatrace, AppDynamics, CA APM, REST/API performance patterns, load/stress/endurance testing, bottleneck analysis, and enterprise release validation. Experienced designing performance test strategies, analyzing response time, throughput, CPU, memory, and network indicators, documenting findings, and guiding engineering teams toward practical tuning and scalability improvements.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built automated Python/Pandas telemetry pipelines ingesting P95 performance and capacity data from 6,000+ endpoints, replacing manual processes with repeatable analytics and reporting workflows.
- Analyzed infrastructure, application, and capacity metrics to identify utilization risk, peak patterns, performance bottlenecks, and optimization opportunities across enterprise banking systems.
- Integrated BMC TrueSight/TSCO, CA APM, and AppDynamics data into unified Oracle reporting datasets for capacity, performance, and executive-level operational visibility.
- Designed optimized Oracle schemas and SQL reporting structures to support historical retention, performance trend analysis, forecasting, and operational risk review.
- Developed Prophet and scikit-learn forecasting models to predict capacity bottlenecks months ahead and support proactive remediation planning.
- Produced dashboards and reports using Streamlit, matplotlib, seaborn, and plotly to communicate performance trends, capacity risk, and recommendations to senior stakeholders.
- Supported root-cause analysis by correlating monitoring data, workload behavior, application performance signals, and infrastructure KPIs.
- Documented repeatable performance reporting processes, validation rules, and operating procedures to improve governance and reduce manual effort.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and other critical systems, supporting production performance monitoring and engineering analysis.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and provide actionable recommendations to engineering teams.
- Integrated HP Performance Center with Dynatrace to correlate load-test behavior, APM traces, infrastructure metrics, and user-experience indicators.
- Implemented End-User Monitoring and end-to-end transaction tracing across web and mobile systems to improve visibility into user journeys and application performance.

- Created performance reports and dashboards that translated telemetry into clear findings for engineering, infrastructure, and stakeholder review.
- Participated in AWS cloud migration readiness discussions by evaluating monitoring coverage, operational risks, and performance data requirements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM subject matter expert for a large financial-services enterprise with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Designed dashboards, alerts, SLA thresholds, and management modules used to monitor enterprise application performance and support operational decision-making.
- Built Perl and Korn Shell data-extraction scripts to automate performance reporting and reduce manual weekly exports.
- Partnered with J2EE/WebLogic and infrastructure teams to diagnose bottlenecks using APM telemetry, JVM indicators, SQL/Oracle signals, thread behavior, and infrastructure metrics.
- Supported large-scale CA APM upgrades and monitoring standardization while maintaining production visibility and minimizing disruption.
- Delivered technical documentation, onboarding guidance, and client training for performance-monitoring standards, agent deployments, and troubleshooting practices.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom web applications under load to identify throughput limits, response-time degradation, and resource bottlenecks including JDBC connections, threads, heap memory, CPU, and garbage collection.
- Used LoadRunner/Performance Center-style performance testing practices to support load, stress, and regression performance validation for enterprise applications.
- Installed and configured JMX Monitoring, Wily Introscope, and thread-dump automation scripts to provide real-time visibility during test execution.
- Documented performance baselines, regression thresholds, findings, and recommendations used by engineering teams for release-gate decisions.
- Collaborated with application, testing, and infrastructure teams to troubleshoot performance defects and validate remediation outcomes.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while maintaining strict performance and reliability expectations.
- Optimized core transaction processing using C++, OCCI, OCI, SQL, and Oracle, reducing physical hardware footprint while preserving sub-second query latency.
- Built C++/OCCI/OCI validation tooling to verify migration correctness and performance-sensitive behavior across large transaction datasets.
- Performed SQL and database-performance analysis to support high-volume transaction processing and migration readiness.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (AMDOCS projects) (2001 – 2007)

- Developed high-availability telecom billing interfaces using C, C++, Pro*C, PL/SQL, Oracle, XML, socket programming, POSIX threads, and Unix environments.

- Improved throughput and database performance in billing processes through SQL tuning, sequence optimization, memory reduction, and interface-level performance analysis.
- Built and maintained integration components using enterprise messaging and interface patterns, including MQ Series and XML-based workflows.
- Automated WebLogic/WebSphere administration and deployment support with Korn Shell scripting, reducing manual operational effort.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt / Cairo, Egypt

Skills

Flagship Projects

Performance Telemetry and Capacity Reporting Platform

Built Python and SQL-driven telemetry reporting workflows to transform enterprise infrastructure and application performance data into dashboards, forecasts, and executive-ready findings.

Technologies: Python, SQL, Oracle, BMC TrueSight, AppDynamics, CA APM, Streamlit

FAST Project

Mined real-user and synthetic monitoring data from Dynatrace and Gomez to identify performance-improvement opportunities for critical revenue-generating systems.

Technologies: Dynatrace, Gomez Synthetic Monitoring, Performance Analytics, Telemetry Reporting

J2EE Performance Diagnostics and Release Validation

Supported load and regression performance validation for J2EE applications using LoadRunner/Performance Center-style practices, JMX monitoring, Wily Introscope, and thread-dump analysis.

Technologies: LoadRunner, Performance Center, J2EE, JMX, Wily Introscope, Thread Dumps

High-Volume Database Migration Validation

Built validation and performance-sensitive migration tooling for a high-throughput travel shopping engine moving from MySQL nodes to Oracle RAC.

Technologies: C++, OCCI, OCI, Oracle RAC, SQL, Performance Validation